Valentin François

Final-year engineering internship
Mechanical design & product development

Availability · Autumn 2027

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Selected projects

Image-based meshing - Python, Gmsh
Image-based triangular mesh generation with refinement driven by image variations.

3D graphics - C, OpenGL
3D geometry visualization: cross-sections and surface analysis using isophotes.

Radio-controlled boat - CAD, CNC, 3D printing
Design and fabrication from plans. Motorized crane with camera feedback.

Experience

Engineering Data · Engineering internship - Fondettes · 2025
Design and sizing of mechanical systems for robotic production lines. Analytical calculations and finite element analysis (SolidWorks Simulation). 3D-printed prototyping, assembly and in-house testing followed by on-site customer trials.

COSTIER STC · Shop-floor internship - Perpignan · 2023
Updated technical documentation and operating and maintenance manuals.

Model building - Association Modélisme Stéphanois · 2010 - present
Design and fabrication of scale models and animated layouts. CAD, CNC milling, laser cutting, 3D printing. Construction of controlled and automated mechanical systems. Delivery of training sessions for association members.

Education

Mechanical engineering · 2022–2027
French engineering degree · UTT

Exchange semester · 2027
Cassino · Italy

Mechanics and in-service performance of materials and products · 2026
Master’s degree · UTT

Exchange in Mechanical Engineering Technology · 2024
Cégep de Sherbrooke · Canada

Aeronautical initiation certificate (BIA) · 2020
Awarded with merit

Skills

CAD: Creo, SolidWorks · Numerical simulation: Abaqus, SolidWorks Simulation
Mechanical sizing · Python, C · Turning, milling · Prototyping

English B2+ · Spanish B1 · Driving licence B
Portfolio · https://valentin-francois.github.io/?lang=en